FCPC: A Dissimilarity-Aware Plug-and-Play Federated Learning Framework for Non-IID IoT Systems

FCPC: 一种用于非独立同分布物联网系统的差异感知即插即用联邦学习框架

Abstract-The proliferation of dual-skewed nonindependent and identically distributed (non-IID) data across Internet of Things (IoT) edge devices—characterized by heterogeneous label distributions and imbalanced sample sizes-severely degrades the robustness of conventional federated learning (FL) frameworks. To address this challenge, we propose FCPC, a dissimilarity-aware plug-and-play FL framework. First, we design a novel metric, Jensen-Shannon Divergence and Skew Indicator Number (JSDN), to quantify the degree of data heterogeneity in edge FL systems. Leveraging JSDN, we develop a client pairing algorithm to select client pairs with maximized distribution dissimilarities. Concurrently, we embed a lightweight regularization term into local training to align gradient update directions across paired clients and mitigate dimensional collapse. Extensive experiments on benchmark datasets (CIFAR-10/100, TinylmageNet, MNIST, FEMNIST) and real-world IoT scenarios validate two key advantages of FCPC: 1) It outperforms state-of-the-art baselines by 2%-9.4% under strong heterogeneity (63.17% top-1 accuracy on CIFAR-10 with $\alpha = 0.05$,9.4 percentage points higher than FedAvg) with negligible computational overhead; 2) As a plug-and-play module, it seamlessly integrates into existing FL algorithms (e.g., FedAvg, FedProx) without modifying their core architectures, yielding a 3.5%-9.4% performance improvement. Overall, this work presents a deployable and compatible solution to enhance FL robustness for practical IoT systems.

摘要——双偏斜非独立同分布 (non-IID) 数据在物联网 (IoT) 边缘设备中的激增——其特征是异构标签分布和不平衡样本大小——严重降低了传统联邦学习 (FL) 框架的鲁棒性。为应对这一挑战，我们提出了 FCPC，一种感知差异的即插即用 FL 框架。首先，我们设计了一种新颖的度量标准，詹森-香农散度和偏斜指标数 (JSDN)，以量化边缘 FL 系统中的数据异质性程度。利用 JSDN, 我们开发了一种客户端配对算法，以选择分布差异最大化的客户端对。同时，我们在局部训练中嵌入一个轻量级正则化项, 以对齐配对客户端之间的梯度更新方向, 并减轻维度坍缩。在基准数据集 (CIFAR-10/100、TinylmageNet、MNIST、FEMNIST) 和实际物联网场景上进行的大量实验验证了 FCPC 的两个关键优势:1) 在强异质性下 (在 CIFAR-10 上，$\alpha = 0.05$ 时 top-1 准确率为 63.17%，比 FedAvg 高 9.4 个百分点)，它以可忽略不计的计算开销比现有最先进的基线高出 2%-9.4%；2) 作为一个即插即用模块，它可以无缝集成到现有 FL 算法 (如 FedAvg、FedProx) 中，而无需修改其核心架构，性能提高 3.5%-9.4%。总体而言，这项工作提出了一种可部署且兼容的解决方案，以增强实际物联网系统的 FL 鲁棒性。

Index Terms- Federated learning(FL), Non-IID data, IoT systems, JSDN, Dissimilarity-Aware, Plug-and-Play, FCPC framework

关键词——联邦学习 (FL)、非 IID 数据、物联网系统、JSDN、差异感知、即插即用、FCPC 框架

I. INTRODUCTION

一、引言

IN the field of distributed machine learning for Internet of Things (IoT) systems, federated learning (FL) has emerged as a pivotal paradigm. It enables collaborative model training across resource-constrained edge devices while preserving data privacy-a critical requirement for sensitive IoT data (e.g., user behavior logs and industrial operational metrics) [1] and a key enabler for the embodied intelligence that interacts with physical IoT environments [2]. FL has been widely applied in IoT applications, including smart healthcare monitoring, industrial predictive maintenance, and intelligent transportation systems. However, data heterogeneity in IoT environments presents a formidable challenge.

在物联网 (IoT) 系统的分布式机器学习领域，联邦学习 (FL) 已成为一种关键范式。它能够在资源受限的边缘设备间进行协作式模型训练，同时保护数据隐私——这是敏感物联网数据 (例如用户行为日志和工业运营指标) 的一项关键要求 N，也是与实体物联网环境交互的具身智能的关键推动因素 [2]。联邦学习已广泛应用于物联网应用，包括智能医疗监测、工业预测性维护和智能交通系统。然而，物联网环境中的数据异质性带来了巨大挑战。

A typical example is the traffic and environmental sensing scenario (Fig. 1), where on-board units, traffic signal controllers, surveillance cameras, and unmanned aerial vehicles (UAVs) generate data characterized by distinct category skew and quantity imbalance (e.g., traffic cameras at urban intersections producing far more samples than those in suburban areas). Such heterogeneity poses notable challenges to latency-sensitive tasks like real-time traffic state prediction and signal timing optimization. Beyond traffic IoT systems, similar data heterogeneity issues persist in other critical industrial tasks reliant on data-driven inference, including load assessment of cylindrical rolling element bearings [3], fault diagnosis in industrial devices via a neuro-fuzzy approach with hypergraph convolution [4], and vibration-based gear wear monitoring and prediction techniques [5]-all of which necessitate robust federated learning frameworks capable of handling non-IID data.

一个典型的例子是交通和环境感知场景 (图 1)，其中车载单元、交通信号控制器、监控摄像头和无人机 (UAV) 生成的数据具有明显的类别偏斜和数量不平衡特征 (例如，城市十字路口的交通摄像头产生的样本比郊区的多得多)。这种异质性给实时交通状态预测和信号定时优化等对延迟敏感的任务带来了显著挑战。除了交通物联网系统，类似的数据异质性问题在其他依赖数据驱动推理的关键工业任务中也存在，包括圆柱滚动轴承的负载评估 [3]、通过超图卷积的神经模糊方法进行工业设备故障诊断 [4] 以及基于振动的齿轮磨损监测和预测技术 [5]——所有这些都需要能够处理非 IID 数据的强大联邦学习框架。

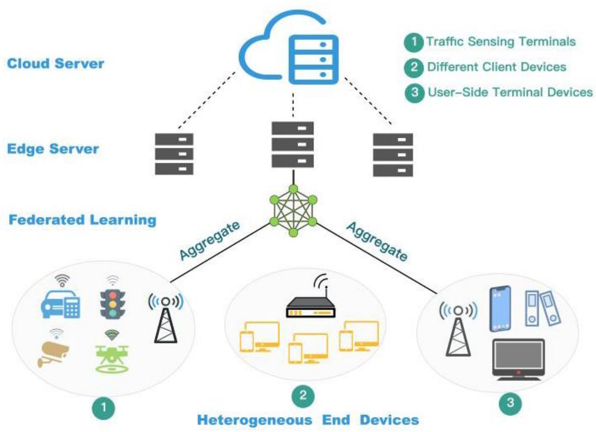

Fig. 1. The non-IID IoT system adopts a cloud-edge-end hierarchical device architecture, which comprises heterogeneous end devices.

图 1. 非 IID 物联网系统采用云-边-端分层设备架构，由异构终端设备组成。

Previous research on federated learning has primarily tackled data heterogeneity from three perspectives, each exhibiting limitations in IoT edge scenarios: Firstly, Algorithmic refinements focus on optimizing local training processes [6], [7] or redesigning global aggregation mechanisms [8]. However, many of these approaches impose substantial computational or communication overhead, which is often incompatible with the resource limitations of typical IoT edge hardware. Secondly, Architectural innovations explore advanced model structures (e.g., Vision Transformers [9]) or novel learning paradigms (e.g., SplitFed [10]) to enhance performance. Nevertheless, such methods frequently depend on specific, non-standardized model architectures or require fundamental changes to the federated learning workflow, hindering their seamless adoption in heterogeneous IoT ecosystems. Thirdly, Temporal-spatial explorations investigate patterns like seasonal data fluctuations [11], yet they may not adequately address the static, structural heterogeneity inherent in many IoT deployments, such as persistent label and quantity skews across devices. A critical gap remains: most existing solutions are tailored to mitigate a single type of data skew and lack effective mechanisms to handle the compounded challenges of dual-skewed non-IID data, where both label distribution and sample size vary significantly across clients-a common scenario in real-world IoT systems.

先前关于联邦学习的研究主要从三个角度解决数据异质性问题，每个角度在物联网边缘场景中都存在局限性: 首先，算法改进侧重于优化局部训练过程 [6]、[7] 或重新设计全局聚合机制 [8]。然而，这些方法中的许多都带来了大量的计算或通信开销，这通常与典型物联网边缘硬件的资源限制不兼容。其次，架构创新探索先进的模型结构 (如视觉 Transformer[9]) 或新颖的学习范式 (如 SplitFed[10]) 以提高性能。然而，这些方法通常依赖于特定的、非标准化的模型架构，或者需要对联邦学习工作流程进行根本性改变，这阻碍了它们在异构物联网生态系统中的无缝采用。第三，时空探索研究季节性数据波动等模式 [11]，但它们可能无法充分解决许多物联网部署中固有的静态、结构异质性，例如设备之间持续的标签和数量偏斜。一个关键差距仍然存在: 大多数现有解决方案都是为减轻单一类型的数据偏斜而量身定制的，缺乏有效机制来处理双偏斜非 IID 数据的复合挑战，即标签分布和样本大小在客户端之间都有显著差异——这是实际物联网系统中的常见情况。

To overcome these limitations, we introduce the Federated Client Pairing and Mutual Constraint (FCPC) framework, motivated by a key observation (detailed in Section 4: Motivation): a potential mutual constraint exists among clients and its strength correlates positively with data distribution dissimilarity. This insight shifts the core challenge to accurately quantifying inter-client data dissimilarity and devising an efficient mechanism to exploit the constraint. FCPC innovates in two aspects. First, it proposes the Jensen-Shannon Divergence and Skew Indicator Number (JSDN) metric, a novel metric that jointly quantifies label and quantity skews. Second, it employs a lightweight JSDN-based client pairing algorithm and injects a pairwise regularization term into local training, actively aligning gradient updates between dissimilar clients. Consequently, FCPC serves as a plug-and-play module that requires no changes to the base FL architecture or aggregation scheme, introduces minimal computational overhead, and maintains orthogonal compatibility-effectively addressing a critical deployment gap in resource-constrained IoT systems.

为克服这些限制，我们引入了联邦客户端配对与相互约束 (FCPC) 框架，其灵感来自于一个关键观察结果 (在第 4 节: 动机中有详细阐述): 客户端之间存在潜在的相互约束，其强度与数据分布差异呈正相关。这一见解将核心挑战转移到准确量化客户端间的数据差异，并设计一种有效的机制来利用这种约束。FCPC 在两个方面进行了创新。首先，它提出了詹森 - 香农散度和偏斜指标数 (JSDN) 度量，这是一种联合量化标签和数量偏斜的新颖度量。其次，它采用了一种基于 JSDN 的轻量级客户端配对算法，并在局部训练中注入成对正则化项，积极使不同客户端之间的梯度更新对齐。因此，FCPC 作为一个即插即用模块，无需对基础联邦学习架构或聚合方案进行更改，引入的计算开销最小，并保持正交兼容性，有效地解决了资源受限的物联网系统中的关键部署差距。

Contributions. The main contributions of this work are summarized as follows:

贡献。这项工作的主要贡献总结如下:

- Novel Insight into Client Constraints: A mutual constraint relationship is identified among federated learning clients, with its strength correlating positively with data distribution dissimilarity. A lightweight pairwise regularization mechanism is designed to leverage this relationship, mitigating dimensional collapse in non-IID settings-particularly crucial for IoT edge systems.

- 对客户端约束的新颖见解: 在联邦学习客户端之间识别出一种相互约束关系，其强度与数据分布差异呈正相关。设计了一种轻量级的成对正则化机制来利用这种关系，减轻非独立同分布 (non-IID) 设置中的维度坍缩，这对物联网边缘系统尤为关键。

- Unified Heterogeneity Quantification Metric: The JSDN metric is proposed to quantify label and quantity skews within a unified formulation. Providing a more holistic characterization of dual-skewed heterogeneity than single-type metrics, JSDN enables more precise client analysis and pairing.

- 统一的异质性量化度量: 提出了 JSDN 度量，以在统一的公式中量化标签和数量偏斜。与单类型度量相比，JSDN 能够更全面地表征双偏斜异质性，从而实现更精确的客户端分析和配对。

- Plug-and-Play Framework for Practical Deployment: We propose the FCPC framework, which does not require core FL algorithms modification. Its design requires no changes to core FL algorithms, operating as an add-on module that introduces only a client pairing step and a lightweight regularization term, thereby enabling practical deployment in heterogeneous IoT environments with minimal integration cost.

- 用于实际部署的即插即用框架: 我们提出了 FCPC 框架，该框架不需要修改核心联邦学习算法。其设计无需对核心联邦学习算法进行更改，作为一个附加模块运行，仅引入一个客户端配对步骤和一个轻量级正则化项，从而能够以最小的集成成本在异构物联网环境中进行实际部署。

II. RELATED WORK

二、相关工作

A. Non-IID IoT Systems

A. 非独立同分布的物联网系统

The data heterogeneity in federated learning refers to the inconsistent distribution of data across edge devices, violating the independent and identically distributed (IID) assumption [12]. In IoT ecosystems, this phenomenon intensifies due to device-specific deployment environments and sensing capabilities. Current heterogeneity classification encompasses two dimensions:

联邦学习中的数据异质性是指跨边缘设备的数据分布不一致, 违反了独立同分布 (IID) 假设 [12]。在物联网生态系统中，由于特定于设备的部署环境和传感能力，这种现象会加剧。当前的异质性分类包括两个维度:

At the label level, it is manifested as differences in the label space of the data from each participant, which exerts a profound impact on model training [13]. Divergent label spaces across devices degrade model robustness. Industrial IoT deployments exemplify this issue, where vibration sensors in high-risk manufacturing zones report disproportionate ”equipment failur” labels, while identical sensors in controlled environments predominantly yield ”normal operation” labels.

在标签层面，它表现为来自每个参与者的数据的标签空间差异，这对模型训练有深远影响 [13]。跨设备的不同标签空间会降低模型的鲁棒性。工业物联网部署就是这个问题的例证，在高风险制造区域的振动传感器报告了不成比例的“设备故障”标签，而在受控环境中的相同传感器主要产生“正常运行”标签。

At the quantity level, this manifests as notable disparities in sample sizes across participants: a single client may possess a vastly larger dataset than the rest, or data volumes may be severely imbalanced among multiple clients.

在数量层面，这表现为参与者之间样本大小的显著差异: 单个客户端可能拥有比其他客户端大得多的数据集，或者多个客户端之间的数据量可能严重不平衡。

B. Existing Solutions

B. 现有解决方案

To contextualize our approach, we review prevalent strategies for handling non-IID data and analyze their suitability for dual-skewed IoT scenarios.

为了将我们的方法置于背景中，我们回顾了处理非独立同分布数据的流行策略，并分析了它们对双偏斜物联网场景的适用性。

FedProx [14] is an approach designed to address system heterogeneity and data heterogeneity. It enables flexible local computation, which to some extent mitigates the interference caused by resource-constrained federated participants during the training process. At the same time, it merges a proximal term in the loss function to avoid excessive deviation of client training from the global model.

FedProx [14] 是一种旨在解决系统异质性和数据异质性的方法。它实现了灵活的局部计算，在一定程度上减轻了训练过程中资源受限的联邦参与者造成的干扰。同时，它在损失函数中合并了一个近端项，以避免客户端训练与全局模型过度偏离。

MOON(Model-Contrastive Learning) [15] mitigates the effects of differences in data distribution by introducing comparative learning techniques into the training process. This method is simple and effective, encouraging model-level feature alignment to improve the performance of federated deep learning models on non-IID datasets.

MOON(模型对比学习)[15] 通过将对比学习技术引入训练过程来减轻数据分布差异的影响。这种方法简单有效，鼓励模型级别的特征对齐，以提高联邦深度学习模型在非独立同分布数据集上的性能。

FastPPO [16] proposes an improved PPO-based scheduling algorithm for personalized federated learning in green IoT. By integrating greedy reward allocation and exponential-decay training duration, it adapts to participants' dynamic resources and preferences, maximizing FL model accuracy while optimizing energy efficiency and participation willingness.

FastPPO [16] 提出了一种改进的基于近端策略优化 (PPO) 的调度算法，用于绿色物联网中的个性化联邦学习。通过整合贪婪奖励分配和指数衰减训练持续时间，它适应参与者的动态资源和偏好，在优化能源效率和参与意愿的同时最大化联邦学习模型的准确性。

FedDyn [17] introduces a new approach to construct agent datasets and dynamically extract local knowledge. Instead of using an averaging strategy, it implements focal distillation to emphasize reliable knowledge, effectively solving the problem of knowledge bias.

FedDyn [17] 引入了一种构建代理数据集并动态提取局部知识的新方法。它不使用平均策略，而是实施焦点蒸馏以强调可靠知识，有效地解决了知识偏差问题。

Summary and Differentiation. As reviewed, the prevailing methods tackle non-IID challenges through diverse strategies: global-model-centric regularization (e.g., FedProx), representation-level contrastive learning (e.g., MOON), dynamic knowledge distillation (e.g., FedDyn), trust evaluation for federated learning in mobile network digital twins (e.g., TFL-DT [18]), local graph-based aggregation for dual skews (e.g., FBLG [19]), causal inference to address aggregation bias (e.g., FedCFA [20]) and reinforcement learning-based personalized scheduling (e.g., FastPPO). While effective in their specific designs, many incur non-negligible computational overhead (e.g., contrastive pairs in MOON, graph construction in FBLG) or require modifications to the core FL aggregation protocol (e.g., VerifyDFL [21], FedDyn, FedCFA), which can hinder their adoption in lightweight, standardized IoT frameworks. Furthermore, except for FBLG which explicitly targets dual skews, most are primarily designed for or evaluated on single-type heterogeneity and lack an intrinsic mechanism to actively leverage inter-client data dissimilarity. In contrast, our proposed FCPC framework is distinguished by its core philosophy of transforming dissimilarity into a convergent force. Specifically, FCPC differs fundamentally by: (1) introducing a lightweight, pairwise constraint that directly utilizes quantified data distribution gaps to align client updates, (2) operating as a plug-and-play module that requires no alteration to the global aggregation scheme and adds minimal computational burden, and (3) employing a unified metric (JSDN) to explicitly target and exploit dual-skewed heterogeneity. This design makes FCPC a uniquely practical and efficient solution tailored for resource-constrained IoT systems where statistical heterogeneity and device limitations coexist.

总结与区分。如前所述，现有方法通过多种策略应对非独立同分布挑战: 以全局模型为中心的正则化 (例如，FedProx)、表示级对比学习 (例如，MOON)、动态知识蒸馏 (例如，FedDyn)、移动网络数字双胞胎中联邦学习的信任评估 (例如，TFL-DT [18])、基于局部图的双偏斜聚合 (例如，FBLG [19])、解决聚合偏差的因果推断 (例如，FedCFA [20]) 以及基于强化学习的个性化调度 (例如，FastPPO)。虽然这些方法在其特定设计中有效，但许多方法会产生不可忽视的计算开销 (例如，MOON 中的对比对、FBLG 中的图构建)，或者需要修改核心联邦学习聚合协议 (例如，VerifyDFL [21]、FedDyn、FedCFA)，这可能会阻碍它们在轻量级、标准化物联网框架中的应用。此外，除了明确针对双偏斜的 FBLG 之外，大多数方法主要是为单类型异质性设计或评估的，并且缺乏主动利用客户端间数据差异的内在机制。相比之下，我们提出的 FCPC 框架的独特之处在于其将差异转化为收敛力的核心理念。具体而言，FCPC 在以下方面存在根本差异:(1) 引入轻量级的成对约束，直接利用量化的数据分布差距来对齐客户端更新；(2) 作为即插即用模块运行，无需更改全局聚合方案，并且增加的计算负担最小；(3) 采用统一度量 (JSDN) 来明确针对和利用双偏斜异质性。这种设计使 FCPC 成为针对统计异质性和设备限制共存的资源受限物联网系统量身定制的独特实用且高效的解决方案。

III.THE SYSTEM MODEL AND DEFINITIONS

三、系统模型与定义

A. Federated Learning Framework

A. 联邦学习框架

As a novel machine learning paradigm, federated learning addresses the challenge of training high-performance global models without disclosing local data when multi-source datasets are distributed across different devices or clients. This framework enables model training on decentralized devices (clients) that maintain local datasets without requiring centralized data exchange.

作为一种新颖的机器学习范式，联邦学习解决了在多源数据集分布于不同设备或客户端时，在不披露本地数据的情况下训练高性能全局模型的挑战。该框架允许在维护本地数据集的分散设备 (客户端) 上进行模型训练，而无需集中式数据交换。

Formally, let $\mathcal{D} = \{D_1, D_2, \dots, D_n\}$ represent the distributed datasets across n clients, where D_i is the local dataset of client i . The objective of federated learning is to train a global model M by aggregating locally computed model updates while minimizing the loss function $\mathcal{L}$ with respect to model parameters θ . This is expressed mathematically as:

形式上，设 $\mathcal{D} = \{D_1, D_2, \dots, D_n\}$ 表示跨 n 个客户端的分布式数据集，其中 D_i 是客户端 i 的本地数据集。联邦学习的目标是通过聚合本地计算的模型更新来训练全局模型 M ，同时相对于模型参数 θ 最小化损失函数 $\mathcal{L}$ 。这在数学上表示为:

$$\min_{\theta} \mathcal{L}(\theta) = \sum_{i=1}^{n} \frac{|D_i|}{|D|} \mathcal{L}_i(\theta), \tag{1}$$

where $|D_i|$ denotes the number of samples in client i ' s dataset, $|D| = \sum_{i=1}^{n} |D_i|$ represents the total number of samples across all clients, and $\mathcal{L}_i$ is the loss function computed on the i -th client' s dataset. This formulation enables effective feature learning from distributed data while ensuring local data residency and privacy preservation.

其中 $|D_i|$ 表示客户端 i 数据集中的样本数量，$|D| = \sum_{i=1}^{n} |D_i|$ 表示所有客户端样本的总数，$\mathcal{L}_i$ 是在第 i 个客户端数据集上计算的损失函数。这种公式化使得能够从分布式数据中进行有效的特征学习，同时确保本地数据的驻留性和隐私保护。

FedAvg [22] represents a foundational and widely deployed algorithm in federated learning. In FedAvg:

FedAvg [22] 是联邦学习中一种基础且广泛应用的算法。在 FedAvg 中:

1) Each client computes model updates based on local data;

1) 每个客户端根据本地数据计算模型更新;

2) Client updates are transmitted to the central server;

2) 客户端更新被传输到中央服务器;

3) The server aggregates updates via weighted averaging:

3) 服务器通过加权平均聚合更新:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \sum_{i=1}^{n} \frac{|D_i|}{|D|} \nabla \mathcal{L}_i\left(\theta^{(t)}\right). \tag{2}$$

Where $\theta^{(t)}$ and $\theta^{(t+1)}$ denote the global model parameters updated at time t and after the update, respectively, and η denotes the learning rate that controls the magnitude of the update.

其中 $\theta^{(t)}$ 和 $\theta^{(t+1)}$ 分别表示在时间 t 更新时和更新后更新的全局模型参数，η 表示控制更新幅度的学习率。

B. Differential Privacy

B. 差分隐私

Differential privacy (DP) [23] provides a rigorous mathematical foundation for privacy preservation in our framework. A randomized mechanism $\mathcal{M}$ satisfies ϵ -differential privacy if: for any two adjacent datasets D and D' differing by at most one element, and any measurable output set $S \subseteq \text{Range}(\mathcal{M})$:

差分隐私 (DP)[23] 为我们框架中的隐私保护提供了严格的数学基础。如果随机机制 $\mathcal{M}$ 满足 ϵ -差分隐私,则对于任何两个最多相差一个元素的相邻数据集 D 和 D' ,以及任何可测输出集 $S \subseteq \text{Range}(\mathcal{M})$:

$$\Pr[\mathcal{M}(D) \in S] \leq e^{\epsilon} \cdot \Pr[\mathcal{M}(D') \in S] \tag{3}$$

where $\epsilon > 0$ is the privacy budget governing the privacy-utility tradeoff. The ℓ_1 -sensitivity of a function $f : \mathcal{D} \to \mathbb{R}^k$ is defined as:

其中 $\epsilon > 0$ 是控制隐私 - 效用权衡的隐私预算。函数 $f : \mathcal{D} \to \mathbb{R}^k$ 的 ℓ_1 -敏感度定义为:

$$\Delta_1 f = \max_{D \sim D'} \left\| f(D) - f(D') \right\|_1 \tag{4}$$

where the maximum is taken over all adjacent datasets. The Laplace mechanism achieves ϵ -DP by adding sensitivity-scaled noise: $\mathcal{M}(D) = f(D) + \text{Lap}(\Delta_1 f/\epsilon)$.

其中最大值是在所有相邻数据集中选取的。拉普拉斯机制通过添加敏感度缩放噪声实现 ϵ -差分隐私: $\mathcal{M}(D) = f(D) + \text{Lap}(\Delta_1 f/\epsilon)$ 。

Extending this to federated systems, local differential privacy (LDP) [24] enforces privacy directly at the data source. A mechanism $\mathcal{M}$ satisfies ϵ -LDP if, for all pairs of inputs $v, v' \in \mathcal{D}$ and any output $y \in \mathcal{Y}$:

将此扩展到联邦系统，局部差分隐私 (LDP)[24] 直接在数据源处强制实施隐私。如果对于所有输入对 $v, v' \in \mathcal{D}$ 和任何输出 $y \in \mathcal{Y}$ ，机制 $\mathcal{M}$ 满足 ϵ -LDP:

$$\frac{\Pr[\mathcal{M}(v) = y]}{\Pr[\mathcal{M}(v') = y]} \leq e^{\epsilon}. \tag{5}$$

This client-side protection eliminates server trust assumptions, aligning with federated learning's core principle of data locality.

这种客户端保护消除了对服务器的信任假设，符合联邦学习的数据局部性核心原则。

In our context, consider client k reporting a label distribution vector $h_k \in \mathbb{N}^C$ (where C is the number of classes) and data size $N_k \in \mathbb{N}$. The ℓ_1 -sensitivities are $\Delta_1 h_k = 2$ (since changing one sample can affect at most two class counts, e.g., decreasing one count and increasing another) and $\Delta_1 N_k = 1$ (as a single sample change alters the size by ± 1). The LDP perturbation mechanisms are:

在我们的场景中，考虑客户端 k 报告一个标签分布向量 $h_k \in \mathbb{N}^C$ (其中 C 是类别数量) 和数据大小 $N_k \in \mathbb{N}$ 。 ℓ_1 -敏感度为 $\Delta_1 h_k = 2$ (因为更改一个样本最多可影响两个类别计数，例如，减少一个计数并增加另一个) 和 $\Delta_1 N_k = 1$ (因为单个样本更改会使大小改变 ± 1)。LDP 扰动机制为:

$$\widetilde{h}_k = h_k + (\text{Lap}(2/\epsilon_h))^C, \tag{6}$$

$$\widetilde{N}_k = N_k + \text{Lap}(1/\epsilon_N), \tag{7}$$

where $\epsilon_h > 0$ and $\epsilon_N > 0$ denote the privacy budgets allocated to the label distribution and size, respectively.

其中 $\epsilon_h > 0$ 和 $\epsilon_N > 0$ 分别表示分配给标签分布和大小的隐私预算。

Through sequential composition, the joint mechanism satisfies ϵ -LDP with $\epsilon = \epsilon_h + \epsilon_N$. To minimize the total variance, the optimal privacy budget allocation is:

通过顺序组合，联合机制满足 ϵ -LDP，隐私预算为 $\epsilon = \epsilon_h + \epsilon_N$ 。为了最小化总方差，最优隐私预算分配为:

$$\epsilon_h^* = \epsilon \cdot \frac{\sqrt{C/N_k}}{\sqrt{C/N_k} + 1}, \ \epsilon_N^* = \epsilon - \epsilon_h^*, \tag{8}$$

where C is the class count and N_k is the local data size. This allocation balances utility between the histogram and scalar protections.

其中 C 是类别计数，N_k 是本地数据大小。这种分配在直方图和标量保护之间平衡了效用。

IV. MOTIVATION

四、动机

A. Dual Skewed Non-IID Data Leads to Worse Global Model Performance Degradation

A. 双偏斜非独立同分布数据导致更严重的全局模型性能下降

We conduct a preliminary experiment with the classic Fe-dAvg algorithm as an example. We set up 10 clients, record the classification accuracy in three scenarios in the number of 20 global rounds, and draw a line graph, as shown in Fig. 2. It can be observed that a single label distribution skew or quantity skew has been able to roughly learn the data characteristics in the previous rounds of communication.

我们以经典的 FedAvg 算法为例进行了初步实验。我们设置了 10 个客户端，在 20 轮全局通信的三种场景下记录分类准确率，并绘制折线图，如图 2 所示。可以观察到，单个标签分布偏斜或数量偏斜在前几轮通信中已经能够大致学习数据特征。

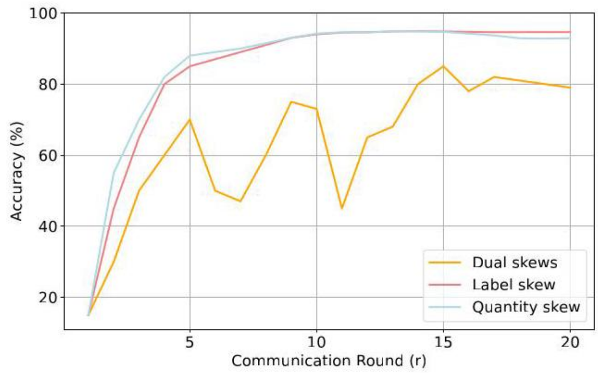

Fig. 2. Impact of dual skewed non-IID data on global model training induced by heterogeneous label distributions among clients and sample size.

图 2. 客户端之间异构标签分布和样本大小引起的双偏斜非独立同分布数据对全局模型训练的影响。

Implication 1. Compared with the single skew, the dual skewed non-IID data has a greater impact on the performance of the global model.

结论 1. 与单偏斜相比，双偏斜非独立同分布数据对全局模型的性能有更大影响。

B. Deep Understanding of Performance Degradation

B. 对性能下降的深入理解

In federated learning, the differences in data distributions across different clients can lead to dimensional collapse in both global and local models, making it difficult for the model to fully utilize the representation space to distinguish between different classes of data, which in turn affects the performance. Implication 2. A bottom-up perspective of dimensional collapse may help minimize the differences in model updates between participants.

在联邦学习中，不同客户端的数据分布差异会导致全局和局部模型中的维度坍缩，使得模型难以充分利用表示空间来区分不同类别的数据，进而影响性能。结论 2. 从自底向上的角度看待维度坍缩可能有助于最小化参与者之间模型更新的差异。

C. Discover Potential Constraints Among Participants

C. 发现参与者之间的潜在约束

Based on the above study, we consider whether heterogeneous clients can restrict each other to mitigate the low ranking of local models during training. To this end, we conducted simple preliminary experiments. Specifically, we first divided the training samples into 10 slices, each corresponding to the local data of one client. We performed a simple experiment on the probability vector $\mathbf{pc} = (\mathrm{pc}_1, \mathrm{pc}_2, \ldots, \mathrm{pc}_K) \sim \mathrm{Dirichlet}(\alpha)$

and assigned the proportion of pc_k of class c instances to clients $k \in \{1, 2, \ldots, K\}$, with $\alpha = 0.01$ to simulate a scenario close to the data heterogeneity in Fig. 3.

基于上述研究，我们考虑异构客户端是否可以相互限制，以减轻训练期间局部模型的低排名。为此，我们进行了简单的初步实验。具体来说，我们首先将训练样本分成 10 个切片，每个切片对应一个客户端的本地数据。我们对概率向量 $\mathbf{pc} = (pc_1, pc_2, \ldots, pc_K) \sim \text{Dirichlet}(\alpha)$ 进行了一个简单实验，并将类别 c 实例的 pc_k 比例分配给客户端 $k \in \{1, 2, \ldots, K\}$，用 $\alpha = 0.01$ 来模拟接近图 3 中数据异构性的场景。

Implication 3. There appears to be a potential constraint effect that can mitigate the dimensional collapse of local models, and this constraint effect is more pronounced among clients with large differences in data distribution. So we need to quantify these differences precisely.

含义 3。似乎存在一种潜在的约束效应，可以减轻局部模型的维度坍缩，并且这种约束效应在数据分布差异较大的客户端中更为明显。因此，我们需要精确量化这些差异。

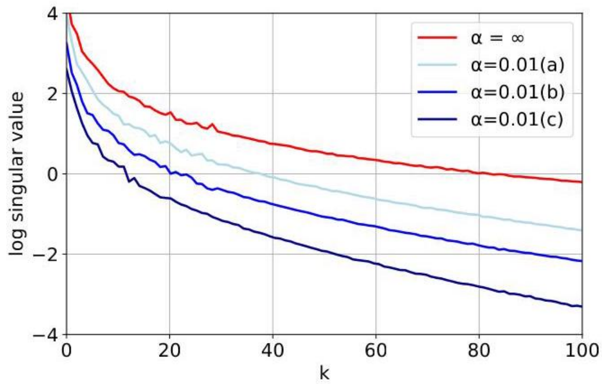

Fig. 3. X-axis(k) is the index of the singular values, and y-axis is the logarithm of the singular values. The red lines represent isomorphic scenarios, and the three lines a, b, c represent paired clients with small and relatively large differences and unpaired, respectively.

图 3。X 轴 (k) 是奇异值的索引，Y 轴是奇异值的对数。红线表示同构场景，三条线 a、b、c 分别表示差异较小、相对较大的配对客户端和未配对客户端。

D. Promoting Alignment of Model Update Directions Among Participants

D. 促进参与者之间模型更新方向的对齐

To counteract the effects of non-IID data, FedDecorr [25] adds a regularization term to the loss function of each local model to encourage tail singularities not to collapse to 0, thus effectively mitigating dimensional collapse.

为了抵消非 IID 数据的影响，FedDecorr [25] 在每个局部模型的损失函数中添加了一个正则化项，以鼓励尾部奇异值不坍缩到 0，从而有效地减轻维度坍缩。

Implication 4. Inspired by FedDecorr, we propose to optimize the training process by integrating a pairwise-based regularization term. The term acts as a mutual restriction on participants with different data distributions, inducing them to try to maintain the same gradient update direction.

含义 4。受 FedDecorr 的启发，我们建议通过集成基于成对的正则化项来优化训练过程。该项对具有不同数据分布的参与者起到相互约束的作用，促使他们尝试保持相同的梯度更新方向。

V. Proposed Method

V. 提出的方法

A. Quantify Data Distribution Differences

A. 量化数据分布差异

After recognizing that the key categories of data heterogeneity are label skew and quantity skew, we design a quantitative metric, JSDN. Specifically, this metric consists of two parts: $\mathcal{JSD}$, which indicates the degree of label skew, and N , which indicates the degree of quantity skew.

在认识到数据异质性的关键类别是标签偏差和数量偏差之后，我们设计了一个定量指标 JSDN。具体来说，这个指标由两部分组成: $\mathcal{JSD}$ ，表示标签偏差的程度，以及 N ，表示数量偏差的程度。

Jensen-Shannon divergence ($\mathcal{JSD}$) is a measure of similarity between two probability distributions [26]. In the federated learning scenario, we choose this metric to measure the similarity of data labeling distributions between different clients. The $\mathcal{JSD}$ value is negatively correlated with the similarity of data distributions, $\mathcal{JSD} \in [0,1]$. In general, the data distribution similarity between clients is inversely proportional to the $\mathcal{JSD}$ value. In the extreme case, when the value is 0, it represents that the two sets of data distributions are identical. The value between two distributions (Qa) and (Qb) can be formally defined as:

詹森-香农散度 ($\mathcal{JSD}$) 是衡量两个概率分布之间相似性的指标 [26]。在联邦学习场景中，我们选择这个指标来衡量不同客户端之间数据标签分布的相似性。$\mathcal{JSD}$ 值与数据分布的相似性呈负相关，$\mathcal{JSD} \in [0,1]$ 。一般来说，客户端之间的数据分布相似性与 $\mathcal{JSD}$ 值成反比。在极端情况下，当值为 0 时，表示两组数据分布相同。两个分布 (Qa) 和 (Qb) 之间的值可以正式定义为:

$$\mathcal{JSD}(Q_a, Q_b) = \frac{1}{2} \sum_{i \in \{a,b\}} \mathcal{KL}(Q_i, \mathbb{M}_{ab}), \tag{9}$$

where Q_a, Q_b are the probability distributions of the data under client a and client b, $\mathcal{KL}(\cdot)$ denotes the KLD (Kullback-Leibler Divergence) [27]. $\mathbb{M}_{ab}$ denotes the mean distribution of Q_a and Q_b :

其中 Q_a, Q_b 是客户端 a 和客户端 b, $\mathcal{KL}(\cdot)$ 下数据的概率分布,表示 KLD(库尔贝克-莱布勒散度)[27]。$\mathbb{M}_{ab}$ 表示 Q_a 和 Q_b 的平均分布:

$$\mathbb{M}_{ab} = \frac{Q_a + Q_b}{2}. \tag{10}$$

N is a quantity skew indicator, representing the difference in data volume distribution among clients. When making predictions on new data, the model may not be able to utilize the knowledge learned from other clients well, leading to a decrease in prediction performance. The data quantity difference between client a and b can be quantified by the following normalized indicator:

N 是一个数量偏差指标，表示客户端之间数据量分布的差异。在对新数据进行预测时，模型可能无法很好地利用从其他客户端学到的知识，导致预测性能下降。客户端 a 和 b 之间的数据量差异可以通过以下归一化指标来量化:

$$N = \frac{|N_a - N_b|}{N_a + N_b}, \tag{11}$$

where N_a and N_b are the data volumes of clients a and b , respectively. This equation calculates the normalized value of the difference in data volume between two clients, with $N \in [0, 1]$. The equation for JSDN is given by:

其中 N_a 和 N_b 分别是客户端 a 和 b 的数据量。这个公式计算两个客户端之间数据量差异的归一化值，$N \in [0, 1]$ 。JSDN 的公式如下:

$$JSDN = (1 - \lambda)\mathcal{JSD} + \lambda N \tag{12}$$

In Eq.(12), λ is a weighting parameter that allows us to adjust the influence of data volume skew relative to label skew. This adjustment is essential, as label skew and volume skew typically have different degrees of impact on the federated learning system's performance. By tuning λ , we can balance the contributions of these two factors in the overall measure of data heterogeneity.

在公式 (12) 中，λ 是一个加权参数，它允许我们调整数据量偏差相对于标签偏差的影响。这种调整是必不可少的，因为标签偏差和量偏差通常对联邦学习系统的性能有不同程度的影响。通过调整 λ ，我们可以在数据异质性的整体度量中平衡这两个因素的贡献。

B. Client Pairing Algorithm

B. 客户端配对算法

1) Data Upload and JSDN Matrix Generation: We assume n clients participate in federated training, each with a dataset corresponding to category C . Before local training begins, each client will first add data perturbations to the local label distribution and data size through LDP, and then upload the processed results to the server. Based on the differential privacy, we have determined a equation that integrates privacy and performance coefficients. The privacy budget allocated to label distribution and data size will be adaptively

adjusted according to n and C . The input for the corresponding JSDN calculation becomes the LDP processed $\hat{\mathbf{p}}$ and $\hat{N}$. Specifically, for client i and client j , the calculated JSDN value is denoted as J_{ij} . Combining all the J_{ij} together forms an n -dimensional JSDN matrix. This matrix provides the basis for the subsequent client pairing.

1) 数据上传与 JSDN 矩阵生成: 我们假设 n 个客户端参与联邦训练，每个客户端都有一个对应类别 C 的数据集。在本地训练开始之前，每个客户端将首先通过局部差分隐私 (LDP) 对本地标签分布和数据大小添加数据扰动，然后将处理后的结果上传到服务器。基于差分隐私，我们确定了一个整合隐私和性能系数的方程。分配给标签分布和数据大小的隐私预算将根据 n 和 C 进行自适应调整。相应 JSDN 计算的输入变为经过 LDP 处理的 $\hat{\mathbf{p}}$ 和 $\hat{N}$ 。具体来说，对于客户端 i 和客户端 j ，计算得到的 JSDN 值记为 J_{ij} 。将所有的 J_{ij} 组合在一起形成一个 n 维的 JSDN 矩阵。这个矩阵为后续的客户端配对提供了基础。

2) Client Pairing Process: When the number of clients n is even, we pair all clients into $n/2$ pairs, starting from the original JSDN matrix that records the JSDN values between all client pairs. We first traverse the JSDN matrix to identify its maximum value $J_{\max}$, with the corresponding row and column subscripts denoted as i and j . This indicates that the data distribution difference between client i and client j is the largest across all pairs. Subsequently, we set all elements in the i -th row, j -th row, i -th column and j -th column of the matrix (excluding the diagonal elements) to zero. For the updated matrix, we again identify the maximum JSDN value and pair the corresponding clients as the second group. This process is repeated iteratively: after each pairing operation, the corresponding rows and columns of the paired clients in the matrix are zeroed out, until all clients are paired. Finally, we obtain $n/2$ client pairs in total. The core idea of this pairing method is to prioritize pairing clients with large data distribution differences together. This is because the constraint relationship between clients with greater distribution differences tends to be stronger.

2) 客户端配对过程: 当客户端数量 n 为偶数时，我们将所有客户端配对成 $n/2$ 对，从记录所有客户端对之间 JSDN 值的原始 JSDN 矩阵开始。我们首先遍历 JSDN 矩阵以确定其最大值 $J_{\max}$ ，其对应的行下标和列下标分别记为 i 和 j 。这表明在所有对中，客户端 i 和客户端 j 之间的数据分布差异最大。随后，我们将矩阵中第 i 行、第 j 行、第 i 列和第 j 列的所有元素 (不包括对角元素) 设置为零。对于更新后的矩阵，我们再次确定最大的 JSDN 值，并将对应的客户端配对作为第二组。这个过程迭代重复: 在每次配对操作之后，矩阵中已配对客户端的相应行和列被清零，直到所有客户端都被配对。最后，我们总共得到 $n/2$ 个客户端对。这种配对方法的核心思想是优先将数据分布差异大的客户端配对在一起。这是因为分布差异较大的客户端之间的约束关系往往更强。

Algorithm 1: Client Pairing

算法 1: 客户端配对

Input : Label distributions $\hat{\mathbf{p}} = \{\hat{\mathbf{p}}_1, \dots, \hat{\mathbf{p}}_n\}$, Data

输入: 标签分布 $\hat{\mathbf{p}} = \{\hat{\mathbf{p}}_1, \dots, \hat{\mathbf{p}}_n\}$ ，数据

sizes $\hat{N} = \left\{\hat{N}_1, \dots, \hat{N}_n\right\}$, Clients n

大小 $\hat{N}=\left\{\hat{N}_1,\ldots,\hat{N}_n\right\}$，客户端 n

Output: Pairing set P_{pairs}

输出: 配对集 P_{pairs}

Initialize $J \leftarrow \text{zeros}\,(n \times n)$

初始化 $J \leftarrow \text{zeros}\,(n \times n)$

for $i \leftarrow 1$ to n do

对于从 $i \leftarrow 1$ 到 n 执行

for $j \leftarrow i+1$ to n do

对于从 $j \leftarrow i+1$ 到 n 执行

$J_{ij} = J_{ji} = \text{JSDN}\left(\hat{\mathbf{p}}_i, \hat{\mathbf{p}}_j; \hat{N}_i, \hat{N}_j\right)$

end

结束

end

结束

$P_{\text{pairs}} \leftarrow \varnothing$

if n even then

如果 n 为偶数，则

for $k \leftarrow 1$ to $n/2$ do

对于从 $k \leftarrow 1$ 到 $n/2$ 执行

$(i^*, j^*) \leftarrow \arg\max J_{ij}, P_{\text{pairs}} \cup = \{(i^*, j^*)\}$

for $x \leftarrow 1$ to n do

对于从 $x \leftarrow 1$ 到 n 执行

$J_{i^*x} = J_{j^*x} = J_{xi^*} = J_{xj^*} = 0$

end

结束

end

结束

else

否则

for $k \leftarrow 1$ to $(n-1)/2$ do

对于从 $k \leftarrow 1$ 到 $(n-1)/2$ 执行

$(i^*, j^*) \leftarrow \arg\max J_{ij}, P_{\text{pairs}} \cup = \{(i^*, j^*)\}$
for $x \leftarrow 1$ to n do

对于从 $x \leftarrow 1$ 到 n 执行

$J_{i^*x} = J_{j^*x} = J_{xi^*} = J_{xj^*} = 0$
end

结束

end

结束

end

结束

return P_{pairs}

返回 P_{pairs}

When n is odd, we can only form $(n-1)/2$ pairs of clients, and there will be one client left at the end. In this case, the minimum value in the JSDN matrix represents that the data distribution difference between this client and other clients is relatively small. From the perspective of data distribution, this client is roughly in the middle position of the client distribution. That is to say, its label distribution and data volume are representative to a certain extent within the overall client cohort, and its discrepancies from other clients are less pronounced than those between certain other client pairs. Therefore, we choose to abandon the client corresponding to the minimum value in the JSDN matrix to avoid the interference of the parity of the number of federated learning participants on the pairing process, thus simplifying the pairing process and improving the adaptability of the pairing algorithm.

当 n 为奇数时，我们只能形成 $(n-1)/2$ 对客户端，最后会剩下一个客户端。在这种情况下，JSDN 矩阵中的最小值表示该客户端与其他客户端之间的数据分布差异相对较小。从数据分布的角度来看，这个客户端大致处于客户端分布的中间位置。也就是说，它的标签分布和数据量在整个客户端群体中在一定程度上具有代表性，并且它与其他客户端的差异比某些其他客户端对之间的差异不太明显。因此，我们选择舍弃 JSDN 矩阵中最小值对应的客户端，以避免联邦学习参与者数量奇偶性对配对过程的干扰，从而简化配对过程并提高配对算法的适应性。

C. FCPC

C. FCPC

Motivated by the above observation and analysis of the potential constraints between heterogeneous clients in federated learning, we explore how to fully utilize such constraints to mitigate the impact of data heterogeneity during federation training. During the original federation training, clients train individually with their local datasets and finally upload the model weights to the central server for federated averaging, during this period, the clients cannot perceive the data distribution gaps with other clients, and even if mutual constraints exist, they cannot achieve the constraint effect. Therefore, we propose to mitigate this problem by including constraints during local training. A natural way to achieve this is to add the following regularization term to the loss function during training. The system architecture is shown in Fig. 4. Formally, the proposed regularization term is defined as:

基于上述对联邦学习中异构客户端之间潜在约束的观察和分析，我们探索如何充分利用这些约束来减轻联邦训练期间数据异构性的影响。在原始的联邦训练中，客户端使用其本地数据集单独训练，最后将模型权重上传到中央服务器进行联邦平均，在此期间，客户端无法感知与其他客户端的数据分布差距，即使存在相互约束，也无法实现约束效果。因此，我们建议通过在本地训练中纳入约束来缓解这个问题。实现这一点的一种自然方法是在训练期间向损失函数添加以下正则化项。系统架构如图 4 所示。形式上，所提出的正则化项定义为:

$$L_{\mathrm{FCPC}}(w_k, X) = \left\| w_k^t - w_{k'}^{t-1} \right\|^2, \tag{13}$$

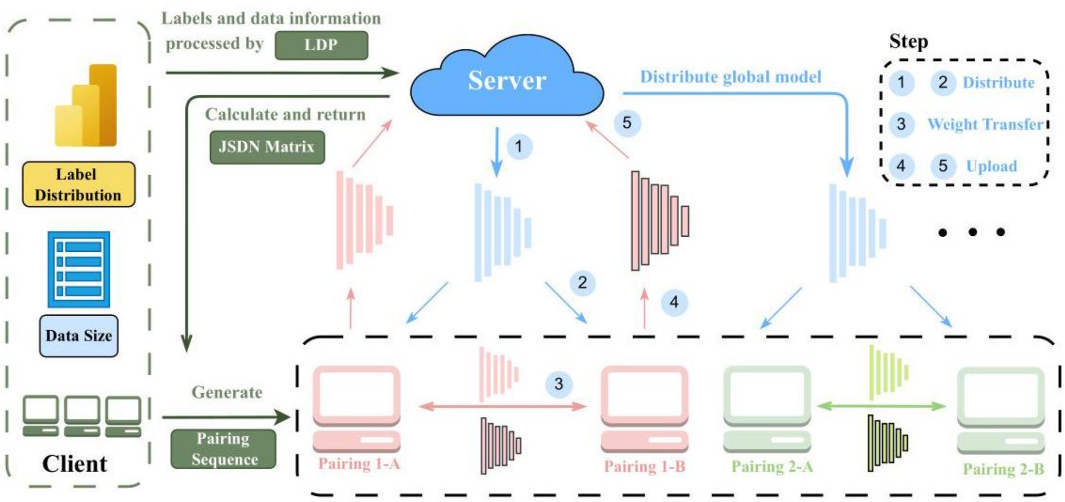

Fig. 4. Framework of FCPC. The left part demonstrates the process that participants in federated learning upload the label distribution and data size to the server after local differential privacy (LDP) processing. The server then calculates the JSDN matrix and returns the pairing sequence. The right part shows the pairing training process of the participants. Taking the red client pair as an example: the server initializes the global model and distributes it to the clients (Steps 1 and 2). At the end of each training round, between the client pairs, the local weights need to be uploaded to the paired client as a constraint term for the client in the next round (Step 3). Meanwhile, the weights are uploaded to the server for global aggregation (Steps 4 and 5).

图 4. FCPC 框架。左半部分展示了联邦学习参与者在进行局部差分隐私 (LDP) 处理后将标签分布和数据大小上传到服务器的过程。服务器然后计算 JSDN 矩阵并返回配对序列。右半部分展示了参与者的配对训练过程。以红色客户端对为例: 服务器初始化全局模型并将其分发给客户端 (步骤 1 和 2)。在每个训练轮次结束时，在客户端对之间，本地权重需要作为下一轮客户端的约束项上传到配对客户端 (步骤 3)。同时，权重被上传到服务器进行全局聚合 (步骤 4 和 5)。

where w_k^t represents the local weight of client k in t -round training and $w_{k'}^{t-1}$ represents the weight of the client paired with client k in the $(t-1)$ -th round of training. At each round, a subset K of the total devices is selected to train. The total communication round is T (As shown in Algorithm 2).

其中 w_k^t 表示客户端 k 在 t 轮训练中的本地权重，$w_{k'}^{t-1}$ 表示在第 $(t-1)$ 轮训练中与客户端 k 配对的客户端的权重。在每一轮中，选择总设备的一个子集 K 进行训练。总通信轮次为 T (如算法 2 所示)。

Each client k is selected with probability p_k (a weight coefficient that quantifies the proportional contribution of its local data to the global model), where $p_k \geq 0$ and $\sum_{k=1}^{n} p_k = 1$. This coefficient is calculated as $p_k = N_k / N_{\text{total}}$, with N_k denoting the data volume of client k and $N_{\text{total}} = \sum_{k=1}^{n} N_k$ representing the total data volume across all n clients.

每个客户端 k 被选中的概率为 p_k (一个权重系数，用于量化其本地数据对全局模型的比例贡献)，其中 $p_k \geq 0$ 且 $\sum_{k=1}^{n} p_k = 1$。该系数计算为 $p_k = N_k / N_{\text{total}}$ ，其中 N_k 表示客户端 k 的数据量，$N_{\text{total}} = \sum_{k=1}^{n} N_k$ 表示所有 n 客户端的总数据量。

After each round of local training, the clients will send the local training weights to the server and the paired clients. The weights sent to the server are used for global aggregation, and the portion sent to the paired clients is used for constructing constraints for the next round of local training for the paired clients. The overall objective of each local client is

在每一轮本地训练之后，客户端会将本地训练权重发送给服务器和配对客户端。发送给服务器的权重用于全局聚合，发送给配对客户端的部分用于为配对客户端的下一轮本地训练构建约束。每个本地客户端的总体目标是

$$\min_{w} \ell(w, X, \mathbf{y}) + \beta L_{\text{FCPC}}(w, X), \tag{14}$$

where ℓ is the cross entropy loss, and β is the regularization coefficient of FCPC. Essentially, L_{FCPC} penalizes the difference between the paired clients, thus preventing the two from going in opposite directions of gradient updating during training. Instead, it constrains the paired clients so that they try to maintain the same direction of gradient updating, speeding up model convergence. This regularization term solves the statistical heterogeneity problem by restricting the local updates of the heterogeneous clients without requiring manually set the number of local epochs. We summarize the steps of FCPC in Algorithm 2.

其中 ℓ 是交叉熵损失，β 是 FCPC 的正则化系数。本质上，L_{FCPC} 惩罚配对客户端之间的差异，从而防止两者在训练期间朝着梯度更新的相反方向发展。相反，它约束配对客户端，使它们尝试保持相同的梯度更新方向，加速模型收敛。这个正则化项通过限制异构客户端的本地更新来解决统计异质性问题，而无需手动设置本地轮数。我们在算法 2 中总结了 FCPC 的步骤。

In addition, detailed convergence analysis of FCPC is provided in the Appendix.

此外，附录中提供了 FCPC 的详细收敛分析。

Algorithm 2: FCPC (Proposed Framework)

算法 2:FCPC(提出的框架)

Input：K, T, β, w^0, n, p_k for $k = 1, \dots, n$

输入：K, T, β, w^0, n, p_k 用于 $k = 1, \dots, n$

Output: Optimized model w^T

输出: 优化后的模型 w^T

for $t \leftarrow 0$ to $T - 1$ do

对于从 $t \leftarrow 0$ 到 $T - 1$ 执行

$S_t \leftarrow$ random subset of K devices

$S_t \leftarrow$ 个 K 设备的随机子集

Send w^t to all $k \in S_t$

将 w^t 发送给所有 $k \in S_t$

foreach device $k \in S_t$ do

对于每个设备 $k \in S_t$ 执行

Find $w_k^{t+1} \approx$

找到 $w_k^{t+1} \approx$

$$\arg\min_{w}\left[\ell\left(w;X,\mathbf{y}\right)+\beta\left\|w_k^t-w_{k'}^{t-1}\right\|^2\right]$$

end

结束

foreach device $k \in S_t$ do

对于每个设备 $k \in S_t$ 执行

Send w_k^{t+1} to server

将 w_k^{t+1} 发送到服务器

end

结束

$w^{t+1} \leftarrow \frac{1}{|S_t|}\sum_{k\in S_t} w_k^{t+1}$

end

结束

return w^T

返回 w^T

VI. Performance Analysis

六、性能分析

A. Experimental Setups

A. 实验设置

In our experimental setup, all participants actively participated in each round of the training process. For network architecture, we used ResNet18 [28] and MobileNetV2 [29], respectively. We performed 100 rounds of communication for all experiments on the CIFAR10/100 [30] datasets. Both CIFAR10 and CIFAR100 have 50,000 training samples and 10,000 test samples, and the size of each image is 32×32 .

在我们的实验设置中，所有参与者都积极参与了每一轮训练过程。对于网络架构，我们分别使用了 ResNet18 [28] 和 MobileNetV2 [29]。我们在 CIFAR10/100 [30] 数据集上对所有实验进行了 100 轮通信。CIFAR10 和 CIFAR100 都有 50,000 个训练样本和 10,000 个测试样本，并且每个图像的大小为 32×32 。

Local training was performed for 10 epochs in each round of communication using the SGD optimizer with a learning rate of 0.01 and a batch size of 64 . In addition, we applied data enhancement to all experiments.

在每一轮通信中，使用学习率为 0.01 且批量大小为 64 的 SGD 优化器进行 10 个轮次的本地训练。此外，我们对所有实验都应用了数据增强。

B. Performance Under Single Types of Non-IID Scenarios

B. 单一类型非 IID 场景下的性能

We selected two small and medium-sized datasets, MNIST [31] and FEMNIST, and carried out a detailed study on the accuracy of the FCPC algorithm in single-type non-IID scenarios. During the experiment, we simulated two common single-type non-IID conditions in Table I: label distribution skew and quantity skew.

我们选择了两个中小型数据集 MNIST [31] 和 FEMNIST，并对 FCPC 算法在单一类型非 IID 场景下的准确性进行了详细研究。在实验过程中，我们模拟了表 I 中的两种常见单一类型非 IID 条件: 标签分布偏差和数量偏差。

For the situation of label distribution skew, the results show that FCPC achieves an accuracy of 0.9815 on the MNIST dataset and an accuracy of 0.9734 on the FEMNIST dataset. In comparison, the performance of other methods on these two datasets is relatively weak. This fully demonstrates that when dealing with data with label distribution skew, FCPC can more effectively mine the feature information in the data, improve the model's recognition ability for different types of data, and thus obtain higher accuracy.

对于标签分布偏差的情况，结果表明 FCPC 在 MNIST 数据集上的准确率为 0.9815，在 FEMNIST 数据集上的准确率为 0.9734。相比之下，其他方法在这两个数据集上的性能相对较弱。这充分表明，在处理具有标签分布偏差的数据时，FCPC 可以更有效地挖掘数据中的特征信息，提高模型对不同类型数据的识别能力，从而获得更高的准确率。

TABLE I

表 I

TEST ACCURACY ON MNIST AND FEMNIST UNDER SINGLE TYPES OF NON-IID SCENARIOS

单一类型非 IID 场景下 MNIST 和 FEMNIST 的测试准确率

Non-IID Scenarios	Dataset	FCPC	FedAvg	FedProx	FedDyn	MOON
Label distribution skew	MNIST	98.15	97.32	97.96	97.48	97.54
	FEMNIST	97.34	96.35	96.28	97.09	96.56
Quantity skew	MNIST	98.28	97.18	97.78	97.63	97.65
	FEMNIST	97.24	95.97	96.37	96.52	96.36

非独立同分布场景	数据集	FCPC	联邦平均	联邦近端算法	联邦动态	MOON
标签分布偏差	MNIST	98.15	97.32	97.96	97.48	97.54
	FEMNIST	97.34	96.35	96.28	97.09	96.56
数量偏差	MNIST	98.28	97.18	97.78	97.63	97.65
	FEMNIST	97.24	95.97	96.37	96.52	96.36

In the experimental scenario of quantity skew, FCPC also demonstrates excellent performance. On the MNIST dataset, the accuracy of FCPC reaches 0.9828, and on the FEMNIST dataset, it is 0.9724 . It can be seen that when facing the situation of large differences in data quantity, FCPC can still maintain a high accuracy, effectively avoiding the problem of model performance degradation caused by data imbalance.

在数量倾斜的实验场景中，FCPC 也表现出优异的性能。在 MNIST 数据集上，FCPC 的准确率达到 0.9828，在 FEMNIST 数据集上，准确率为 0.9724。可以看出，当面对数据量差异较大的情况时，FCPC 仍能保持较高的准确率，有效避免了数据不平衡导致的模型性能下降问题。

C. Performance Under Dual Types of Non-IID Scenarios

C. 双类型非 IID 场景下的性能

Addressing mixed types of data skews (e.g., concurrent label and quantity skews) is recognized as a significantly more challenging task in federated learning than handling single-type skews, as the interplay between different heterogeneities can lead to compounded performance degradation. To evaluate the efficacy of FCPC under such complex conditions, experiments were conducted on the CIFAR-10 dataset under dual-skew scenarios. The degree of heterogeneity was controlled using a Dirichlet distribution parameter $\alpha \in \{0.05, 0.1, 0.5\}$, simulating environments with extreme, high, and moderate data heterogeneity, respectively.

在联邦学习中, 处理混合类型的数据倾斜 (例如并发标签和数量倾斜) 被认为是一项比处理单一类型倾斜更具挑战性的任务，因为不同异质性之间的相互作用可能导致性能下降加剧。为了评估 FCPC 在这种复杂条件下的有效性，在双倾斜场景下对 CIFAR-10 数据集进行了实验。使用狄利克雷分布参数 $\alpha \in \{0.05, 0.1, 0.5\}$ 控制异质性程度，分别模拟了极端、高和中等数据异质性的环境。

Experimental results, summarized in Table II, demonstrate that the proposed FCPC framework consistently achieves superior classification accuracy across all tested heterogeneity levels. In contrast, the performance of all baseline methods exhibits a more substantial decline, particularly under the extreme heterogeneity condition ($\alpha = 0.05$) .

表二总结的实验结果表明，所提出的 FCPC 框架在所有测试的异质性水平上都始终实现了卓越的分类准确率。相比之下，所有基线方法的性能都有更显著的下降，特别是在极端异质性条件 ($\alpha = 0.05$) 下。

TABLE II

表二

PERFORMANCE COMPARISON UNDER DIFFERENT LEVELS OF DUAL-SKEWED NON-IID DATA ON CIFAR-10 AND CIFAR-100

CIFAR-10 和 CIFAR-100 上不同双倾斜非 IID 数据水平下的性能比较

Dataset	CIFAR10			CIFAR100		
Method / α	0.05	0.1	0.5	0.05	0.1	0.5
FedAvg (AISTATS, 2017)	53.74	65.19	79.98	49.76	56.43	61.40
FedProx (MLsys, 2020)	53.21	65.20	79.46	50.22	56.37	61.59
MOON (CVPR, 2021)	59.68	67.60	80.04	45.68	56.37	61.81
FedDyn (ICPADS, 2023)	62.24	68.49	77.38	49.73	56.36	61.28
FBLG (IJCAI, 2024)	62.95	69.13	79.82	50.45	57.62	61.38
FedCFA(AAAI, 2025)	63.15	69.83	80.45	51.47	57.81	62.13
FCPC	63.17	70.59	80.28	52.53	58.12	62.72

数据集	CIFAR10			CIFAR100		
方法 / α	0.05	0.1	0.5	0.05	0.1	0.5
联邦平均算法 (AISTATS，2017 年)	53.74	65.19	79.98	49.76	56.43	61.40
联邦近端算法 (MLsys，2020 年)	53.21	65.20	79.46	50.22	56.37	61.59
MOON(CVPR，2021 年)	59.68	67.60	80.04	45.68	56.37	61.81
联邦动态算法 (ICPADS，2023 年)	62.24	68.49	77.38	49.73	56.36	61.28
FBLG(IJCAI，2024 年)	62.95	69.13	79.82	50.45	57.62	61.38
联邦 CFA(AAAI，2025 年)	63.15	69.83	80.45	51.47	57.81	62.13
FCPC	63.17	70.59	80.28	52.53	58.12	62.72

D. Orthogonality of FCPC with Existing FL Algorithms

D. FCPC 与现有联邦学习算法的正交性

Thanks to the regularization property of FCPC method, it acts on the local training stage through pairing between devices. It only needs to introduce the pairing algorithm on the basis of other existing methods and add an additional term L_{FCPC} on the loss function. In addition, it does not need any change, which makes FCPC have the natural advantage of plug-and-play.

得益于 FCPC 方法的正则化特性，它通过设备间的配对作用于局部训练阶段。只需在其他现有方法的基础上引入配对算法，并在损失函数上添加一个额外的项 L_{FCPC} 。此外，无需任何更改，这使得 FCPC 具有即插即用的天然优势。

To validate this orthogonality, FCPC is integrated into five representative baseline algorithms, namely FedAvg [22], FedProx [14], MOON [15], FBLG [19], and FedCFA [20]. Two sets of experiments are conducted: (1) CIFAR-10 dataset with the ResNet-18 model, where data are partitioned among 10 clients; (2) TinyImageNet dataset with the same client partitioning strategy. The degree of heterogeneity is quantified using a Dirichlet distribution with $\alpha \in \{0.05, 0.1, 0.5\}$ (smaller α indicates stronger heterogeneity). The experimental objective is to verify whether integrating FCPC can enhance the performance of baseline methods under varying non-IID degrees.

为了验证这种正交性，将 FCPC 集成到五种代表性的基线算法中，即 FedAvg [22]、FedProx [14]、MOON [15]、FBLG [19] 和 FedCFA [20]。进行了两组实验:(1) 使用 ResNet-18 模型的 CIFAR-10 数据集，数据在 10 个客户端之间进行划分；(2) 使用相同客户端划分策略的 TinyImageNet 数据集。使用具有 $\alpha \in \{0.05, 0.1, 0.5\}$ 的狄利克雷分布来量化异构程度 (α 越小表示异构性越强)。实验目的是验证集成 FCPC 是否能在不同的非 IID 程度下提高基线方法的性能。

Experimental results indicate that embedding FCPC as a regularization term into baseline algorithms yields notable performance gains, especially under strongly heterogeneous settings ($\alpha = 0.05$ and $\alpha = 0.1$) . On the CIFAR-10 dataset, the performance improvement ranges from approximately 4.0% to 9.4% compared with the original baselines. For instance, FedAvg achieves 53.74% accuracy when $\alpha = 0.05$, and the accuracy increases to 63.17% after integrating FCPC, representing a 9.43% improvement. On the TinyImageNet dataset, the improvement under $\alpha = 0.05$ and $\alpha = 0.1$ is about 5.0% and 3.5%-4.5%, respectively.

实验结果表明，将 FCPC 作为正则化项嵌入到基线算法中能带来显著的性能提升，尤其是在强异构设置 ($\alpha = 0.05$ and $\alpha = 0.1$) 下。在 CIFAR-10 数据集上，与原始基线相比，性能提升范围约为 4.0% 至 9.4%。例如，当 $\alpha = 0.05$ 时，FedAvg 的准确率为 53.74%，集成 FCPC 后准确率提高到 63.17%，提升了 9.43%。在 TinyImageNet 数据集上，在 $\alpha = 0.05$ 和 $\alpha = 0.1$ 下的提升分别约为 5.0% 和 3.5%-4.5%。

The results presented in Tables III and IV collectively demonstrate the effectiveness and orthogonality of integrating FCPC with existing FL algorithms. Notably, FCPC consistently enhances the performance of baseline methods across different datasets and heterogeneity levels, which verifies its ability to strengthen the non-IID data handling capability of FL frameworks.

表 III 和 IV 中的结果共同证明了将 FCPC 与现有联邦学习算法集成的有效性和正交性。值得注意的是，FCPC 在不同数据集和异构水平上持续提高基线方法的性能，这验证了其增强联邦学习框架处理非 IID 数据能力的能力。

TABLE III

表 III

ORTHOGONALITY ANALYSIS OF FCPC INTEGRATION WITH EXISTING FL ALGORITHMS (RESNET-18, CIFAR10)

FCPC 与现有联邦学习算法集成的正交性分析 (ResNet-18, CIFAR10)

α	FedAvg (AISTATS 2017)		FedProx (MLsys 2020)		MOON (CVPR 2021)		FBLG (IJCAI 2024)		FedCFA (AAAI 2025)	
	Orig.	+FCPC	Orig.	+FCPC	Orig.	+FCPC	Orig.	+FCPC	Orig.	+FCPC
0.05	53.74	63.17	53.21	61.29	59.68	63.46	62.95	63.08	63.15	64.23
0.1	65.19	70.59	65.20	72.83	67.60	72.26	69.13	70.45	69.83	72.29
0.5	79.98	80.28	79.46	80.35	80.04	80.82	79.82	80.83	80.45	81.62

α	联邦平均算法 (AISTATS 2017)		联邦近端算法 (MLsys 2020)		MOON(CVPR 2021)		FBLG(IJCAI 2024)		联邦 CFA(AAAI 2025)	
	原始版本	+FCPC	原始版本	+FCPC	原始版本	+FCPC	原始版本	+FCPC	原始版本	+FCPC
0.05	53.74	63.17	53.21	61.29	59.68	63.46	62.95	63.08	63.15	64.23
0.1	65.19	70.59	65.20	72.83	67.60	72.26	69.13	70.45	69.83	72.29
0.5	79.98	80.28	79.46	80.35	80.04	80.82	79.82	80.83	80.45	81.62

TABLE IV

表四

ORTHOGONALITY ANALYSIS OF FCPC INTEGRATION WITH EXISTING FL ALGORITHMS (TINY-IMAGENET)

FCPC 与现有 FL 算法 (TINYIMAGENET) 集成的正交性分析

Method	TinyImageNet		
	$\alpha = 0.05$	$\alpha = 0.1$	$\alpha = 0.5$
FedAvg (AISTATS 2017)	35.48	39.53	46.97
+FCPC	40.47	44.36	50.28
FedProx (MLsys 2020)	35.50	39.69	47.23
+FCPC	40.68	44.33	50.53
MOON (CVPR 2021)	35.49	40.81	47.91
+FCPC	40.64	44.42	51.32
FedCFA (AAAI 2025)	35.62	40.94	48.11
+FCPC	40.71	44.57	51.48

方法	微小图像网		
	$\alpha = 0.05$	$\alpha = 0.1$	$\alpha = 0.5$
联邦平均 (AISTATS 2017)	35.48	39.53	46.97
+FCPC	40.47	44.36	50.28
联邦近端 (MLsys 2020)	35.50	39.69	47.23
+FCPC	40.68	44.33	50.53
MOON(CVPR 2021)	35.49	40.81	47.91
+FCPC	40.64	44.42	51.32
联邦 CFA(AAAI 2025)	35.62	40.94	48.11
+FCPC	40.71	44.57	51.48

E. Ablation Study

E. 消融研究

To deeply explore the impact of the key hyperparameters in the FCPC algorithm and the selection of the pairing indicators of the federated learning participants on the model performance, we carried out comprehensive ablation experiments, focusing on the parameter λ , the parameter β , and the similarity measure. These key parameters play a crucial role in the FCPC algorithm, directly affecting the model's processing effect on data heterogeneity and the final performance.

为深入探究 FCPC 算法中关键超参数以及联邦学习参与者配对指标的选择对模型性能的影响，我们进行了全面的消融实验，重点关注参数 λ、参数 β 和相似性度量。这些关键参数在 FCPC 算法中起着至关重要的作用，直接影响模型对数据异质性的处理效果和最终性能。

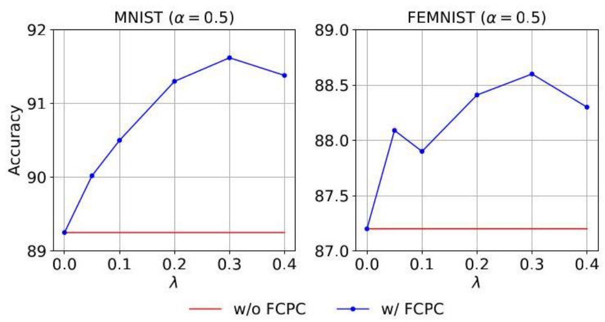

Fig. 5. Ablation studies of λ . We set the value of α to 0.5 and select the MNIST and FEMNIST datasets to conduct experiments with λ taking values from the set $\{0.1, 0.2, 0.3, 0.4\}$ to explore the appropriate value of λ .

图 5. λ 的消融研究。我们将 α 的值设置为 0.5，并选择 MNIST 和 FEMNIST 数据集，让 λ 从集合 $\{0.1, 0.2, 0.3, 0.4\}$ 中取值进行实验，以探索 λ 的合适值。

Coefficient λ . Considering the characteristic of the FedAvg algorithm itself, which performs weighted averaging based on the amount of data, the label skew may play a dominant role. Based on this, we search for the relatively most appropriate value of λ in the set $\{0.1, 0.2, 0.3, 0.4\}$ to measure the distribution difference between clients. As λ gradually increases, the model's adaptability to the data amount difference is enhanced. However, when λ is too large, it may overemphasize the data amount deviation and relatively weaken the impact of the label deviation, causing the model performance to be affected again. After a series of experimental comparisons, it is found that when $\lambda = 0.3$, the JSDN indicator can more effectively measure the distribution difference of client data, enabling the FCPC algorithm to achieve good performance in various experimental scenarios. Meanwhile, we also conducted a rigorous theoretical derivation of λ values in the Appendix. As shown in Fig. 5, it is recommended to set λ to 0.3 to fully utilize the advantages of the FCPC algorithm in handling dual-skewed non-IID data.

系数 λ。考虑到 FedAvg 算法本身基于数据量进行加权平均的特性，标签偏差可能起主导作用。基于此，我们在集合 $\{0.1, 0.2, 0.3, 0.4\}$ 中搜索 λ 的相对最合适值，以衡量客户端之间的分布差异。随着 λ 逐渐增大，模型对数据量差异的适应性增强。然而，当 λ 过大时，可能会过度强调数据量偏差，相对削弱标签偏差的影响，导致模型性能再次受到影响。经过一系列实验比较，发现当 $\lambda = 0.3$ 时，JSDN 指标能够更有效地衡量客户端数据的分布差异，使 FCPC 算法在各种实验场景中都能取得良好性能。同时，我们还在附录中对 λ 的值进行了严格的理论推导。如图 5 所示，建议将 λ 设置为 0.3，以充分利用 FCPC 算法在处理双偏斜非 IID 数据方面的优势。

Coefficient β . We systematically evaluated the impact of the hyperparameter β on FCPC' s performance across various data-heterogeneity scenarios (controlled by $\alpha \in \{0.05, 0.1\}$). Experiments were conducted with $\beta \in \{0.05, 0.1, 0.2, 0.3\}$ while keeping all other conditions (network architecture, dataset, training epochs, optimizer settings, and data augmentation) identical. The results show that FCPC's performance improves with increasing β , stabilizes, and then slightly declines. Moderate β values strengthen regularization, enhancing inter-client gradient consistency and improving performance. Conversely, excessive β introduces overly strong constraints that hinder learning, leading to suboptimal performance. Across all experimental configurations, $\beta = 0.2$ consistently yielded near-optimal results (see Fig. 6), representing an ideal balance between constraint strength and learning capacity. This value serves as a robust default when prior information is unavailable.

系数 β。我们系统地评估了超参数 β 在各种数据异质性场景 (由 $\alpha \in \{0.05, 0.1\}$ 控制) 下对 FCPC 性能的影响。在保持所有其他条件 (网络架构、数据集、训练轮次、优化器设置和数据增强) 相同的情况下，使用 $\beta \in \{0.05, 0.1, 0.2, 0.3\}$ 进行实验。结果表明，FCPC 的性能随着 β 的增加而提高，然后稳定下来，随后略有下降。适度的 β 值会加强正则化，增强客户端间梯度一致性并提高性能。相反，过大的 β 会引入过强的约束，阻碍学习，导致性能次优。在所有实验配置中，$\beta = 0.2$ 始终产生接近最优的结果 (见图 6)，代表了约束强度和学习能力之间的理想平衡。当没有先验信息时，这个值可作为一个可靠的默认值。

TABLE V

表五

PERFORMANCE COMPARISON OF DIFFERENT DISSIMILARITY METRICS AT $\alpha = 0.5$. JSDN REPRESENTS THE COMPREHENSIVE INDICATOR OF JENSEN-SHANNON DIVERGENCE AND DATA SIZE.

$\alpha = 0.5$ 下不同差异度量的性能比较。JSDN 代表詹森 - 香农散度和数据大小的综合指标。

Dataset	Dissimilarity Metrics			
	JSDN	Jensen-Shannon divergence	Euclidean distance	Cosine distance
MNIST	91.63	90.52	90.27	90.38
FEMNIST	88.61	87.47	86.30	86.45
CIFAR10	80.08	78.52	76.75	76.70

数据集	差异度量			
	JSDN	詹森 - 香农散度	欧几里得距离	余弦距离
MNIST	91.63	90.52	90.27	90.38
FEMNIST	88.61	87.47	86.30	86.45
CIFAR10	80.08	78.52	76.75	76.70

Dissimilarity measures. We first validate the effect of the JSDN metric adopted in this paper on the performance of FCPC at $\alpha = 0.5$, and then replace the dissimilarity metrics with Jensen-Shannon divergence, cosine distance, and Euclidean distance, respectively, and then present the classification accuracy of the four metrics in a table, as shown in Table V. We observe that our proposed FCPC performs best on the JSDN dissimilarity metric. This preference is attributed to the unique advantage of JSDN to measure both label distribution and quantity distribution.

差异度量。我们首先在 $\alpha = 0.5$ 验证本文采用的 JSDN 度量对 FCPC 性能的影响，然后分别用 Jensen-Shannon 散度、余弦距离和欧几里得距离替换差异度量，然后在表中呈现这四种度量的分类准确率，如表五所示。我们观察到，我们提出的 FCPC 在 JSDN 差异度量上表现最佳。这种偏好归因于 JSDN 在测量标签分布和数量分布方面的独特优势。

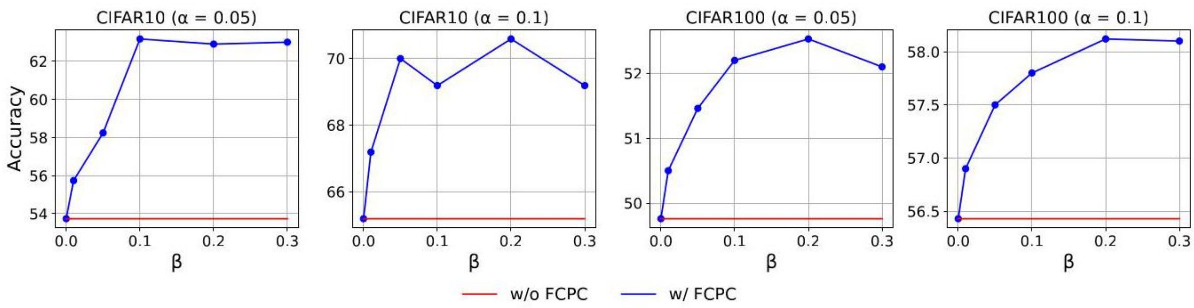

Fig. 6. Ablation studies on β . We select a value of β that is as adaptive as possible to multiple scenarios by comparing the effect of different regularization strengths on model performance on FedAvg.

图 6. 在 β 上的消融研究。我们通过比较不同正则化强度对 FedAvg 上模型性能的影响，选择一个尽可能适应多种场景的 β 值。

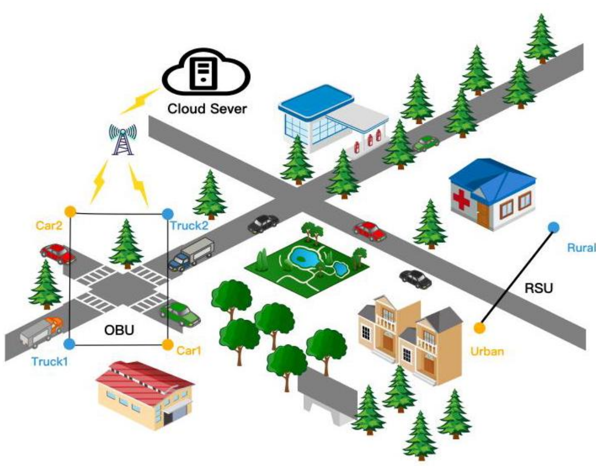

Fig. 7. The heterogeneous information sources in urban and rural areas form the RSU, while different modes of transportation such as trucks and cars form the OBU, collectively forming the heterogeneous data network of the IoT.

图 7. 城乡地区的异构信息源形成 RSU，而卡车和汽车等不同运输方式形成 OBU，共同构成物联网的异构数据网络。

F. Engineering Applications

F. 工程应用

In intelligent transportation systems, the collaborative optimization of traffic signal timing and vehicle route planning relies heavily on the fusion of multi-source IoT data, among which the vehicle-road collaborative network composed of roadside units (RSUs), on-board units (OBUs), and traffic light controllers is a typical application scenario. These IoT devices are distributed in diverse road segments and generate a large amount of non-IID traffic data due to differences in traffic flow density, vehicle types, and travel purposes: for example, RSUs at urban intersections in peak hours mainly collect data with high congestion labels and large sample sizes, while RSUs in suburban roads mostly obtain data with low congestion labels and small sample sizes; OBUs of freight vehicles on highways generate data with stable high-speed characteristics, which are significantly different from the data with frequent start-stop characteristics generated by OBUs of passenger cars in urban areas.

在智能交通系统中，交通信号定时和车辆路线规划的协同优化严重依赖于多源物联网数据的融合，其中由路边单元 (RSU)、车载单元 (OBU) 和交通灯控制器组成的车路协同网络是一个典型的应用场景。这些物联网设备分布在不同的路段，由于交通流密度、车辆类型和出行目的的差异，会产生大量非独立同分布的交通数据: 例如，高峰时段城市路口的 RSU 主要收集高拥堵标签和大样本量的数据，而郊区道路的 RSU 大多获取低拥堵标签和小样本量的数据；高速公路上货运车辆的 OBU 产生具有稳定高速特征的数据，这与城市地区乘用车 OBU 产生的频繁启停特征的数据有显著差异。

To address this, the FCPC algorithm can be integrated into the intelligent transportation IoT system (see Fig. 8): each RSU, OBU, and traffic light controller acts as a client in federated learning, and first uploads the label distribution and data size of local traffic data to the cloud server after LDP processing; the server calculates the JSDN matrix based on the uploaded information, and pairs clients with significant differences in data distribution through the client pairing algorithm; during local training, the paired clients constrain each other through the FCPC regularization term, which reduces the divergence of gradient update directions caused by non-IID data, enables the local models to learn the characteristics of heterogeneous traffic data more comprehensively, and then aggregates the local model parameters to form a global model. This global model can be used to predict short-term traffic flow in each road segment and optimize traffic signal timing dynamically, thereby reducing traffic congestion and improving travel efficiency.

为了解决这个问题，可以将 FCPC 算法集成到智能交通物联网系统中 (见图 8): 每个 RSU、OBU 和交通灯控制器在联邦学习中充当客户端，首先在进行局部差分隐私 (LDP) 处理后将本地交通数据的标签分布和数据大小上传到云服务器；服务器根据上传的信息计算 JSDN 矩阵，并通过客户端配对算法将数据分布差异显著的客户端进行配对；在局部训练期间，配对的客户端通过 FCPC 正则化项相互约束，这减少了由非独立同分布数据引起的梯度更新方向的差异，使局部模型能够更全面地学习异构交通数据的特征，然后聚合局部模型参数形成全局模型。这个全局模型可用于预测每个路段的短期交通流量并动态优化交通信号定时，从而减少交通拥堵并提高出行效率。

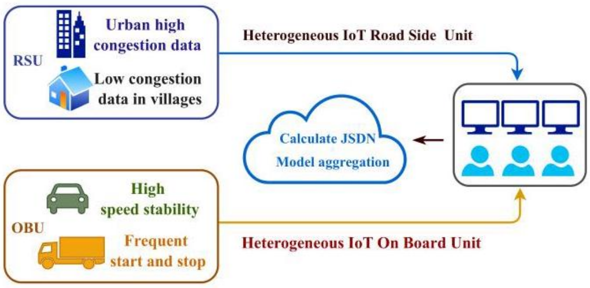

Fig. 8. FCPC in IoT. This diagram illustrates the engineering application logic of the FCPC framework in IoT-driven intelligent transportation systems.

图 8. 物联网中的 FCPC。此图说明了 FCPC 框架在物联网驱动的智能交通系统中的工程应用逻辑。

In practical applications, the plug-and-play nature of FCPC allows new IoT devices to be seamlessly integrated into the federated learning system without modifying the underlying framework, which is highly adaptable to the continuous expansion of intelligent transportation infrastructure. Our subsequent work will focus on applying the FCPC algorithm to the collaborative response of unexpected traffic events, exploring how to use the mutual constraint mechanism between heterogeneous clients to improve the real-time performance and accuracy of event handling, and further enhancing the intelligence level of the transportation system.

在实际应用中, FCPC 的即插即用特性允许新的物联网设备无缝集成到联邦学习系统中, 而无需修改底层框架，这对智能交通基础设施的不断扩展具有高度适应性。我们后续的工作将集中于将 FCPC 算法应用于意外交通事件的协同响应, 探索如何利用异构客户端之间的相互约束机制提高事件处理的实时性能和准确性，并进一步提升交通系统的智能水平。

VII. CONCLUSION

VII. 结论

This paper addresses the critical challenge of performance degradation in federated learning (FL) caused by dual-skewed (label and quantity) non-IID data in Internet of Things (IoT) systems. A key insight driving this work is that the mutual constraint potential between clients is positively correlated with their data distribution dissimilarity. Building on this insight, a novel, dissimilarity-aware framework named FCPC is proposed, which comprises two synergistic components: (1) the Jensen-Shannon Divergence and Skew Indicator Number (JSDN) index for precise client heterogeneity quantification, and (2) a client pairing and mutual-constraint mechanism that employs a lightweight regularization term during local training to align gradient updates and mitigate dimensional collapse.

本文解决了物联网 (IoT) 系统中由双偏斜 (标签和数量) 非独立同分布数据导致的联邦学习 (FL) 性能下降这一关键挑战。推动这项工作的一个关键见解是，客户端之间的相互约束潜力与其数据分布差异呈正相关。基于这一见解，提出了一种名为 FCPC 的新颖的、基于差异感知的框架，它由两个协同组件组成:(1) 用于精确量化客户端异构性的 Jensen-Shannon 散度和偏斜指标数 (JSDN) 指数，以及 (2) 一种客户端配对和相互约束机制，该机制在局部训练期间采用轻量级正则化项来对齐梯度更新并减轻维度坍缩。

Extensive experiments on multiple datasets (CIFAR-10/100, TinyImageNet, MNIST, FEMNIST) and network architectures (ResNet-18, MobileNetV2) confirm FCPC's efficacy. It consistently outperforms strong baselines (FedAvg, FedProx, MOON), achieving top-1 accuracy on CIFAR-10 under severe dual-skew conditions. Crucially, FCPC's plug-and-play design enables seamless integration into existing FL frameworks without introducing prohibitive computational burden, offering a practical system integration strategy for resource-constrained IoT environments.

在多个数据集 (CIFAR-10/100、TinyImageNet、MNIST、FEMNIST) 和网络架构 (ResNet-18、MobileNetV2) 上进行的大量实验证实了 FCPC 的有效性。它始终优于强大的基线 (FedAvg、FedProx、MOON)，在严重双偏斜条件下在 CIFAR-10 上实现了 top-1 准确率。至关重要的是，FCPC 的即插即用设计能够无缝集成到现有的联邦学习框架中，而不会带来过高的计算负担，为资源受限的物联网环境提供了一种实用的系统集成策略。

Future research will explore: (1) adaptive mechanisms for dynamic parameter tuning in evolving IoT environments; (2) extension to broader heterogeneity types (e.g., feature skew and concept drift); and (3) validation in cross-silo industrial FL scenarios with stringent privacy requirements.

未来的研究将探索:(1) 在不断发展的物联网环境中进行动态参数调整的自适应机制；(2) 扩展到更广泛的异构类型 (例如，特征偏差和概念漂移)；以及 (3) 在具有严格隐私要求的跨孤岛工业联邦学习场景中进行验证。

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